

Jaeha Kim

其他

基本信息

姓名: Jaeha Kim

所属院校: 首尔国立大学

地区: South Korea

海外经历: 有

学科领域

微电子科学与工程

集成电路设计与集成系统

研究领域

模拟与数模混合电路

混合信号集成技术

模拟与射频IC设计

射频集成电路

超大规模集成电路设计

低功耗电路与能量管理

EDA工具与电路验证

通信系统与网络 (通信工程)

数字集成电路设计

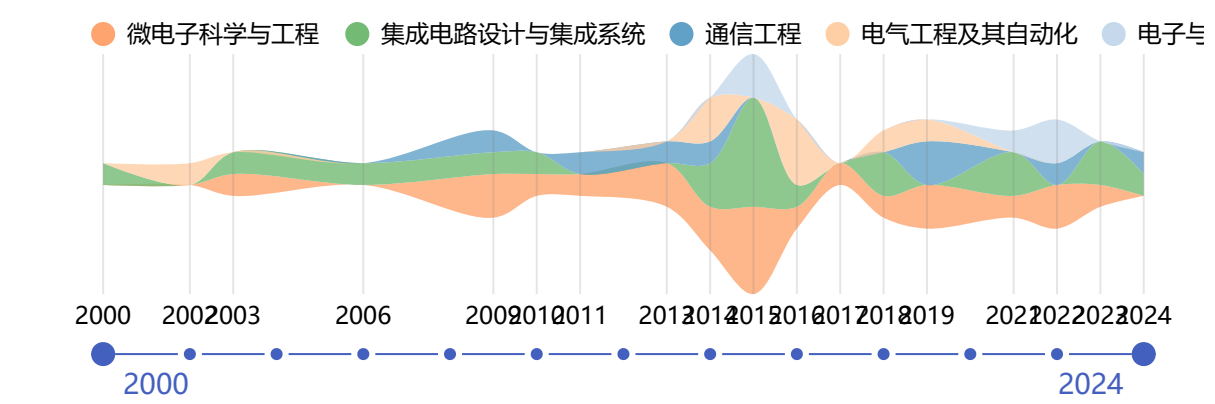
个人简介

Jaeha Kim is a distinguished professor in the Department of Electrical and Computer Engineering at Seoul National University, Korea. His research focuses on integrated circuit design and design automation. With a verified email at snu.ac.kr, he has been cited over 4,707 times, holding an h-index of 32. His notable collaborations include work with esteemed institutions such as Stanford University and Harvard University. His contributions to the field are evident through his numerous highly cited publications in areas like optimized power supply systems and self-powered wireless transmission.

工作经历

Seoul National University

近年发表的论文



Fractionally-Spaced Equalizers as Clock and Data Recovery Loops

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Ryu, Sigang; Kim, Jaeha; Raychowdhury, Arijit

被引用次数: 0 2024

A 1.22fJ/cycle 17T Pseudo-Static True Single-Phase Clock Flip-Flop in 14nm FinFET CMOS

2024 50TH IEEE EUROPEAN SOLID-STATE ELECTRONICS RESEARCH CONFERENCE, ESSERC 2024

Lee, Dongkwon; Kim, Sunghoon; Kang, Minyoung; Jang, Seoyoon; Park, Sungjin; Kim, Jaeha; Jeon, Dongsuk

被引用次数: 0 2024

Reachability Analysis for Nonlinear Analog/Mixed-Signal Circuits With Trajectory-Based Reachable Sets

IEEE ACCESS

Kim, Seyoung; Kim, Jaeha

被引用次数: 0 2023

Accurate Layout-Dependent Effect Model in 10 nm-Class DRAM Process Using Area-Efficient Array Test Circuits

IEEE ACCESS

Kim, Seyoung; Yang, Seungho; Lim, Hyein; Lee, Hyein; Jeon, Jongwook; Choi, Jung Yun; Kim, Jaeha

被引用次数: 0 2023

A 2.5-32 Gb/s Gen 5-PCIe Receiver With Multi-Rate CDR Engine and Hybrid DFE

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Choi, Moon-Chul; Lee, Sanghee; Roh, Seungha; Lee, Kwangho; Oh, Jonghyun; Kim, Sungwoo; Kim, Kwandong; Choi, Woo-Seok; Kim, Jaeha; Jeong, Deog-Kyoon

被引用次数: 5 2022

A Quadrature Error Corrector for Aperiodic, Quarter-rate Data Strobe Signals in HBM3 Interfaces

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Jo, Seo-Yeong; Lee, Jinhyung; Park, Myeong-Jae; Jeong, Deog-Kyoon; Kim, Jaeha

被引用次数: 0

2022

XSNN: a System-Level Simulator for Spiking Neural Network with Neuron Circuits and Synapse Devices

2022 19TH INTERNATIONAL SOC DESIGN CONFERENCE (ISOC)

Min, Jeong Woo; Kim, Jaeha

被引用次数: 0

2022

A Gradient-Descent Calibration Method to Mitigate Process Variations in Analog Synapse Arrays

2022 INTERNATIONAL CONFERENCE ON ELECTRONICS, INFORMATION, AND COMMUNICATION (ICEIC)

Baek, Seung-Heon; Kim, Jaeha

被引用次数: 2

2022

A Time-Multiplexing Multi-Input Hybrid Harvester with Low-Power MPPT Technique

2021 IEEE INTERNATIONAL CONFERENCE ON CONSUMER ELECTRONICS-ASIA (ICCE-ASIA)

Yu, Seungho; Kim, Byoungkyu; Kim, Jaeha

被引用次数: 0

2021

A Time-Based Pipelined ADC Using Integrate-and-Fire Multiplying-DAC

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Ryu, Sigang; Park, Chan Young; Kim, Wooryeol; Son, Seuk; Kim, Jaeha

被引用次数: 2

2021

Safety Verification of AMS Circuits with Piecewise-Linear System Reachability Analysis

18TH INTERNATIONAL SOC DESIGN CONFERENCE 2021 (ISOC 2021)

Kim, Seyoung; Park, Heechun; Kim, Jaeha

被引用次数: 1

2021

Fast Automatic Circuit Optimization Using Deep Learning

18TH INTERNATIONAL SOC DESIGN CONFERENCE 2021 (ISOC 2021)

Hong, Jiwoo; Kim, Sunghoon; Kim, Jaeha; Jeon, Dongsuk

被引用次数: 0

2021

A 2x Blind Oversampling FSE Receiver With Combined Adaptive Equalization and Infinite-Range Timing Recovery

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Son, Seuk; Ryu, Sigang; Yeo, Hwanseok; Kim, Jaeha

被引用次数: 4

2019

A channel-emulating high-speed transmitter with pseudo-logarithmic and low-bandwidth amplifiers

IEICE ELECTRONICS EXPRESS

Kim, Kyunghoon; Kim, Jaeha

被引用次数: 0

2019

Development of a tunable low-frequency vibration energy harvester and its application to a self-contained wireless fatigue crack detection sensor

STRUCTURAL HEALTH MONITORING-AN INTERNATIONAL JOURNAL

Yang, Suyoung; Jung, Sung-Youb; Kim, Kiyoung; Liu, Peipei; Lee, Sangmin; Kim, Jaeha; Sohn, Hoon

被引用次数: 12

2019

An Accurate and Noise-Resilient Spread-Spectrum Clock Tracking Aid for Digitally-Controlled Clock and Data Recovery Loops

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Ryu, Sigang; Son, Seuk; Kim, Jaeha

被引用次数: 3

2019

CHIMERA: A Field-Programmable Mixed-Signal IC With Time-Domain Configurable Analog Blocks

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Choi, Yunju; Lee, Yoontaek; Baek, Seung-Heon; Lee, Sung-Joon; Kim, Jaeha

被引用次数: 2

2018

Demonstration of Unsupervised Learning With Spike-Timing-Dependent Plasticity Using a TFT-Type NOR Flash Memory Array

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kim, Chul-Heung; Lee, Soochang; Woo, Sung Yun; Kang, Won-Mook; Lim, Suhwan; Bae, Jong-Ho; Kim, Jaeha; Lee, Jong-Ho

被引用次数: 55

2018

Electronic skins for soft, compact, reversible assembly of wirelessly activated fully soft robots

SCIENCE ROBOTICS

Byun, Junghwan; Lee, Yoontaek; Yoon, Jaeyoung; Lee, Byeongmoon; Oh, Eunho; Chung, Seungjun; Lee, Takhee; Cho, Kyu-Jin; Kim, Jaeha; Hong, Yongtaek

被引用次数: 193

2018

Design and Analysis of Energy-Efficient Single-Pulse Piezoelectric Energy Harvester and Power Management IC for Battery-Free Wireless Remote Switch Applications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Lee, Minbok; Yang, Joonseok; Park, Myeong-Jae; Jung, Sung-Youb; Kim, Jaeha

被引用次数: 19

2018

In Vivo Self-Powered Wireless Transmission Using Biocompatible Flexible Energy Harvesters

ADVANCED FUNCTIONAL MATERIALS

Kim, Dong Hyun; Shin, Hong Ju; Lee, Hyunseung; Jeong, Chang Kyu; Park, Hyewon; Hwang, Geon-Tae; Lee, Ho-Yong; Joe, Daniel J.; Han, Jae Hyun; Lee, Seung Hyun; Kim, Jaeha; Joung, Boyoung; Lee, Keon Jae

被引用次数: 182

2017

Introduction to the Special Section on the 2016 Asian Solid-State Circuits Conference (A-SSCC 2016)

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jeong, Deog-Kyoon; Kim, Jaeha

被引用次数: 0

2017

Efficient Global Optimization of Analog Circuits Using Predictive Response Surface Models on Discretized Design Space

IEEE DESIGN & TEST

Lee, Jiho; Kim, Jaeha

被引用次数: 0

2016

Design of a 93% Energy-Efficient Buck-Type Capacitor Charger IC in 250-nm CMOS

IEEE TRANSACTIONS ON INDUSTRY APPLICATIONS

Lee, Minbok; Kim, Jaeha

被引用次数: 9

2016

A 500-MHz, 0.76-W/mm² Power Density and 76.2% Power Efficiency, Fully Integrated Digital Buck Converter in 65-nm CMOS

IEEE TRANSACTIONS ON INDUSTRY APPLICATIONS

Lee, Minbok; Choi, Yunju; Kim, Jaeha

被引用次数: 30

2016

Design and Analysis of 37.5% Energy-Recycling Flyback-Type Class-D Low-Side Gate Driver IC with 5-to-15-V Level Conversion

IEEE TRANSACTIONS ON INDUSTRY APPLICATIONS

Kang, Taewook; Lee, Yoontaek; Kim, Jaeha

被引用次数: 0

2016

PPV-Based Modeling and Event-Driven Simulation of Injection-Locked Oscillators in SystemVerilog

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Jang, Jieun; Kim, Jaeha

被引用次数: 2

2015

Investigations on the Optimal Support Vector Machine Classifiers for Predicting Design Feasibility in Analog Circuit Optimization

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Lee, Jiho; Kim, Jaeha

被引用次数: 0

2015

Digital Analog Design: Enabling Mixed-Signal System Validation

IEEE DESIGN & TEST

Lim, Byong Chan; Mao, James; Horowitz, Mark; Jang, Ji-Eun; Kim, Jaeha

被引用次数: 14

2015

GIDL Characteristics in Gated-Diode Memory String and Its Application to Current-Steering Digital-to-Analog Conversion

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kim, Chul-Heung; Lee, Ju-Wan; Kim, Junseok; Kim, Jaeha; Lee, Jong-Ho

被引用次数: 6

2015

Design of Low-Power and Low-Latency 256-Radix Crossbar Switch Using Hyper-X Network Topology

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Baek, Seung-Heon; Jung, Sung-Youb; Kim, Jaeha

被引用次数: 0

2015

A 9-11-Bit Phase-Interpolating Digital Pulsewidth Modulator With 1000x Frequency Range

IEEE TRANSACTIONS ON INDUSTRY APPLICATIONS

Lee, Yoontaek; Kang, Taewook; Kim, Jaeha

被引用次数: 9

2015

Introduction to the December Special Issue on the 2015 IEEE International Solid-State Circuits Conference

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Takamiya, Makoto; Wu, Jieh-Tsorng; Ryyanen, Jussi; Okada, Kenichi; Kim, Jaeha

被引用次数: 0

2015

A Reconfigurable Rectified Flexible Energy Harvester via Solid-State Single Crystal Grown PMN-PZT

ADVANCED ENERGY MATERIALS

Hwang, Geon-Tae; Yang, Joonseok; Yang, Seong Ho; Lee, Ho-Yong; Lee, Minbok; Park, Dae Yong; Han, Jae Hyun; Lee, Seung Jun; Jeong, Chang Kyu; Kim, Jaeha; Park, Kwi-Il; Lee, Keon Jae

被引用次数: 121

2015

A 9.2 GHz Digital Phase-Locked Loop With Peaking-Free Transfer Function

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Ryu, Sigang; Yeo, Hwanseok; Lee, Yoontaek; Son, Seuk; Kim, Jaeha

被引用次数: 16

2014

A Single-Inductor, Multiple-Channel Current-Balancing LED Driver for Display Backlight Applications

IEEE TRANSACTIONS ON INDUSTRY APPLICATIONS

Kim, Hyun-Chang; Yoon, Chang Soo; Jeong, Deog-Kyoon; Kim, Jaeha

被引用次数: 58

2014

Linearization Technique for Binary Phase Detectors in a Collaborative Timing Recovery Circuit

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Yoo, Byoung-Joo; Bae, Woo-Rham; Han, Jiho; Kim, Jaeha; Jeong, Deog-Kyoon

被引用次数: 4	2014
Variability-Aware, Discrete Optimization for Analog Circuits	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Jung, Seobin; Lee, Jiho; Kim, Jaeha	
被引用次数: 9	2014
Yield-Aware Pareto Front Extraction for Discrete Hierarchical Optimization of Analog Circuits	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Jung, Seobin; Lee, Jiho; Kim, Jaeha	
被引用次数: 15	2014
Maximum energy transfer condition for piezoelectric energy harvesters with single pulsed vibration inputs	
ELECTRONICS LETTERS	
Yang, Joonseok; Lee, Sanggu; Kim, Jaeha	
被引用次数: 10	2014
Pseudo-Linear Analysis of Bang-Bang Controlled Timing Circuits	
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS	
Park, Myeong-Jae; Kim, Jaeha	
被引用次数: 43	2013
A 2.3-mW, 5-Gb/s Low-Power Decision-Feedback Equalizer Receiver Front-End and its Two-Step, Minimum Bit-Error-Rate Adaptation Algorithm	
IEEE JOURNAL OF SOLID-STATE CIRCUITS	
Son, Seuk; Kim, Han-Seok; Park, Myeong-Jae; Kim, Kyunghoon; Chen, E-Hung; Leibowitz, Brian; Kim, Jaeha	
被引用次数: 28	2013
Equalizer Design and Performance Trade-Offs in ADC-Based Serial Links	
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS	
Kim, Jaeha; Chen, E. -Hung; Ren, Jihong; Leibowitz, Brian S.; Satarzadeh, Patrick; Zerbe, Jared L.; Yang, Chih-Kong Ken	
被引用次数: 34	2011
Fast, Non-Monte-Carlo Estimation of Transient Performance Variation Due to Device Mismatch	
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS	
Kim, Jaeha; Jones, Kevin D.; Horowitz, Mark A.	
被引用次数: 13	2010
Simulation and Analysis of Random Decision Errors in Clocked Comparators	
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS	
Kim, Jaeha; Leibowitz, Brian S.; Ren, Jihong; Madden, Chris J.	

被引用次数: 112

2009

Design Optimization of On-Chip Inductive Peaking Structures for 0.13- μm CMOS 40-Gb/s Transmitter Circuits

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Kim, Jaeha; Kim, Jeong-Kyoum; Lee, Bong-Joon; Jeong, Deog-Kyoon

被引用次数: 44

2009

Replica compensated linear regulators for supply-regulated phase-locked loops

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Alon, E; Kim, J; Pamarti, S; Chang, K; Horowitz, M

被引用次数: 134

2006

Self-biased high-bandwidth low-jitter 1-to-4096 multiplier clock generator PLL

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Maneatis, JG; Kim, J; McClatchie, L; Maxey, J; Shankaradas, M

被引用次数: 135

2003

An efficient digital sliding controller for adaptive power-supply regulation

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, J; Horowitz, MA

被引用次数: 77

2002

A variable-frequency parallel I/O interface with adaptive power-supply regulation

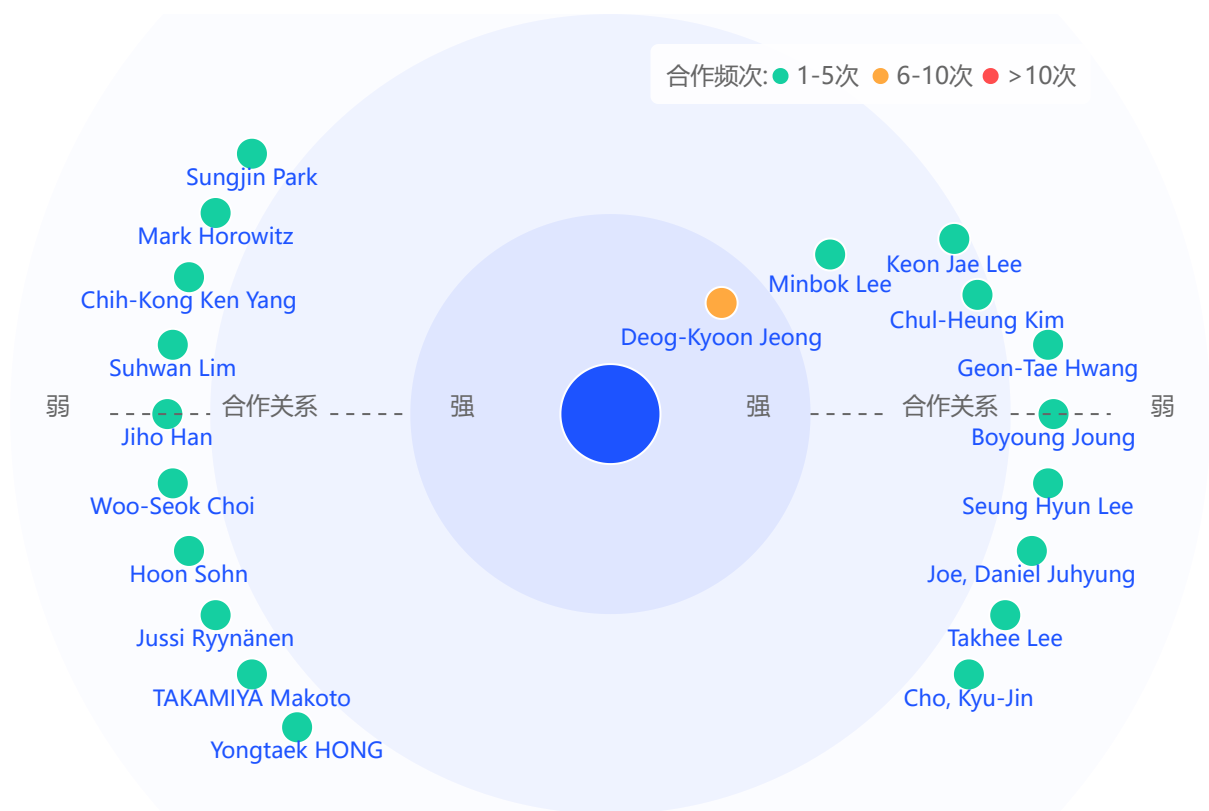
IEEE JOURNAL OF SOLID-STATE CIRCUITS

Wei, GY; Kim, J; Liu, D; Sidiropoulos, S; Horowitz, MA

被引用次数: 92

2000

学术合作



Deog-Kyoon Jeong

合作次数: 6

首尔国立大学

Minbok Lee

合作次数: 4

首尔国立大学

Keon Jae Lee

合作次数: 2

韩国科学技术院

Chul-Heung Kim

合作次数: 2

首尔国立大学

Geon-Tae Hwang

合作次数: 1

韩国科学技术院

还有 15 个合作者